ADITI GUPTA

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Electronics & Telecommunication Engineering graduate (BV(DU)COE, Pune, 2026) with hands-on experience in embedded systems, RF/antenna design, and radar signal processing, including a DRDO (DIAT) research internship. Seeking an entry-level R&D or graduate-engineer role in sensing, embedded systems, and signal processing.

RESEARCH & PROJECT EXPERIENCE

Team Member — IDRO (Research & Innovation Foundation) *Jul 2025 – Present*

- Led design, 3D-printed fabrication, and assembly of IDRO's first prototype drone, from concept through flight-ready build — the foundation's first completed hardware project.
- Served as Technical Advisor, establishing the foundation's open-source presence via its GitHub organization page.

Research Intern — Defence Institute of Advanced Technology (DIAT), DU, DRDO, Pune *May 2025 – Jul 2025*

- Co-developed an unpublished research manuscript on H-slotted microstrip patch antenna design for dual-band RFID applications, simulating −25 dB return loss at 2.58 GHz and 7.6 dBi gain via CST Studio Suite, under the guidance of Dr. Rajesh Singh.
- Completed a 3-month government R&D internship including hardware prototyping, simulation work, and technical reporting.

PUBLICATIONS & AWARDS

- **Best Paper Award (Author)** — RRRAI 2026 International Conference on Responsible, Risk-Aware, and Regulated AI, BVDU COEP — for a paper on wearable health-monitoring technology, selected from international submissions for methodological rigor.

EDUCATION

Bharati Vidyapeeth (Deemed to be University) College of Engineering, Pune *2022 – 2026*
B. Tech, Electronics and Telecommunication Engineering — CGPA: **8.4**

Relevant coursework: Embedded Systems, Digital Signal Processing, VLSI & Digital Logic, Computer Networks, Probability & Statistics, Discrete Mathematics, DBMS

St. Francis Hr. Sec. School, Bilaspur, Chhattisgarh *2019 – 2021*
XII (CBSE): 83.6% | X (CBSE): 80%

TECHNICAL SKILLS

HDLs: Verilog, VHDL
VLSI & Digital Logic: CMOS circuit design, logic gates, flip-flops, adders, multiplexers, synchronous circuits
Embedded Systems: ARM LPC2148, 8051, ESP32 (FreeRTOS); UART, I2C, GSM, LCD interfacing; real-time/RTOS concepts
RF & Signal Processing: Antenna design & simulation (CST Studio Suite), FIR/IIR digital filter design, FMCW radar signal processing
Programming: C, Python, Java, SQL
Tools & Platforms: Vivado, Proteus, Keil µVision, Arduino IDE, MATLAB, Altium Designer, EasyEDA (PCB design), Power BI
Research Tools: LaTeX, Jupyter

KEY PROJECTS

Wearable Multi-Sensor Radar-Based Health Monitoring System — Custom PCB, ESP32, Python (CNN)

- Designed a 2-layer custom PCB (top and bottom) integrating a 24 GHz FMCW radar module (RD-03E) with IMU, biosignal, and temperature sensors for contact and contactless sensing modes.
- Built ESP32/FreeRTOS firmware for real-time multi-sensor data acquisition and a CNN-based classification pipeline, reaching approximately 80% classification accuracy in preliminary, informal testing (not yet formally validated on a defined test set).

Digital Hearing Aid — Python, DSP

- Built a real-time audio-processing application comparing FIR vs. IIR filter architectures for noise reduction, optimizing for sub-100ms latency and minimal phase distortion.

Traffic Light Controller (VHDL FSM) & Semaphore-Based Task Sync on ARM LPC2148 — VHDL, Keil µVision, RTOS

- Designed an FSM-based controller in VHDL; separately used semaphores to synchronize tasks and shared resources on LPC2148, ensuring safe multitasking.

CERTIFICATIONS

- **VLSI SoC Design using Verilog HDL —** Maven Silicon
- **100 Days of Code: The Complete Python Pro Bootcamp** — Udemy
- **Java Masterclass 2025 (130+ hours)** — Udemy

POSITIONS OF RESPONSIBILITY

Editorial & Envoy Head, ETSA — BVDU COEP *Sep 2024 – Aug 2025*

- Grew and managed ETSA's LinkedIn presence and led production of the association's print/digital magazine across multiple editions.

Design Team Member, Google Developer Student Club (GDSC), BVP Pune *Sep 2023 – Jun 2024*

- Designed web pages and UI/UX layouts as a member of the GDSC Design Team, contributing visual and interaction design for club initiatives.